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Structure and Infrastructure Engineering
Maintenance, Management, Life-Cycle Design and Performance

ISSN: 1573-2479 (Print) 1744-8980 (Online) Journal homepage: www.tandfonline.com/journals/nsie20

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To cite this article: Xi Zhu, Tengfei Bao, Justin K. W. Yeoh, Ningxiao Jia & Hui Li (2023) Enhancing dam safety evaluation using dam digital twins, Structure and Infrastructure Engineering, 19:7, 904-920, DOI: 10.1080/15732479.2021.1991387

To link to this article: https://doi.org/10.1080/15732479.2021.1991387

Published online: 20 Oct 2021.

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STRUCTURE AND INFRASTRUCTURE ENGINEERING
2023, VOL. 19, NO. 7, 904–920
https://doi.org/10.1080/15732479.2021.1991387

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Enhancing dam safety evaluation using dam digital twins

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ABSTRACT

Comprehensive evaluation of dams in a dynamic and proactive way is accepted as an effective strategy to improve dam safety. However, there is a lack of efficient and standardised approaches for managing monitoring-related information that can help to provide dynamic and reliable information for continuous dam evaluation. With the importance of digital twins (DTs) being proven in better data integration and interoperability, a DT-based approach for comprehensive dam evaluation and its data integration method based on extended industry foundation classes (IFC) are provided in this study. This paper presents a new data structure and corresponding data classification strategy, using which information required for continuous evaluation can be effectively organised. Considering that existing evaluation methods cannot fully characterise uncertainties in the evaluation process, a cloud model based comprehensive evaluation framework is proposed. In this approach, a multi-rule cloud reasoning model is utilised to evaluate monitoring points. Also, a multi-dimensional cloud model and an improved Criteria Importance Though Intercriteria Correlation (CRITIC) method are leveraged to assess evaluation indices affected by multiple factors. Finally, the results of a case study verify that the proposed DT-based solution realises a continuous dam safety evaluation, which contributes to automated and efficient dam condition monitoring.

ARTICLE HISTORY
Received 19 March 2021
Revised 30 July 2021
Accepted 3 August 2021

KEYWORDS
Digital twins; building information modelling; industry foundation classes; dam monitoring system; long-term monitoring; comprehensive dam evaluation; cloud model

1. Introduction

Dam projects have created huge economic and social benefits for human beings throughout history, especially in water supply, flood control, and renewable energy production (Huang & Yan, 2009; Yüksel, 2010). However, most as-built dams suffer from safety-related issues (e.g., excessive deformation (Chen, Gu, Bao, Wu, & Su, 2016), adverse seepage (Jiang et al., 2020), slope instability (Martt, 2005), etc.) due to the combined influence of many reasons such as increasing natural hazard risks, aging of existing infrastructure, lack of experience in dam safety management and operations, among others. These emerging safety issues may lead to potential dam accidents and further cause adverse impacts on the life, property, and environment of downstream areas, if they are not detected and resolved in time. A comprehensive evaluation of dam service behaviour is the most important and scientific way to monitor dam operation condition and diagnose potential problems (Wu, Xu, Gu, & Li, 2012).

Recently, with the increasing awareness of risk prevention and control in the field of dam engineering, some research has demonstrated the necessity of the transformation of comprehensive dam evaluation mode from the traditional static and passive one to a dynamic and proactive one (Jeon, Lee, Shin, & Park, 2009; Supakchukul et al., 2019; Wu & Su, 2005). Its initiative aims to maintain dynamic information, support continuous condition monitoring of dam structures, achieve efficient decision-making, and finally minimise the risk of dam failure. The new mode of dam safety evaluation is of vital significance for diagnosing potential problems and taking reinforcement measures timely especially when some unpredicted catastrophic floods are coming or a united safety management of dam groups in the same river basin is required (Huang, Li, Hu, Zhang, & Zhao, 2020; Li, Bao, & Fang, 2005).

However, in the traditional static mode, measured data that contains diagnostic information is read from different kinds of instruments or sensors and stored in one or several databases (Li & Wang, 2017), while detailed information (e.g., monitoring precision and frequency, correction parameters and methods) of instruments are hidden in the instruction manuals and location information are presented in the monitoring instrument layout (usually a set of design drawings). The information related to dam safety monitoring is usually saved in different formats and separately stored in different management systems. In view of this, dam engineers may require significant time and effort to query and coordinate all necessary information for evaluation purposes. Hence the current distributed and informal information management status in dam engineering cannot

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efficiently provide dynamic information for continuous safety evaluation.

To support assessment-related tasks, some advanced techniques have been adopted to improve data availability and reliability. For example, using ASP.NET technology, Li and Wang (2017) constructed an integrated monitoring platform based on Brower/Server structure to enhance emergency response operations. In this platform, rapid access to monitoring data is achieved by a data acquisition system. Jancetić, Kranjčić, and Rezo (2020) leveraged SQL (Structural Query Language) and GIS (Geo-Information System) tools in dam management. GIS allows easy storage, processing, and analysis of spatial data to support dam safety management. Although some efforts have been made to integrate multiple information and facilitate data utilisation. The massive information is still stored and described in an informal manner. It may lead to misunderstanding of various monitoring data sources and further leave room for error, because complex information cross-references are quite frequent during comprehensive evaluation process.

Building information modelling (BIM) or referred to as 'digital twin' (DT), aiming to digitise and informatize various physical and functional characteristics of a real entity, has brought new ideas to assets condition monitoring (Kaewunruen, Sresakoolchai, Ma, & Phil-Ebosie, 2021; Lu, Xie, Parlikad, & Schooling, 2020) in recent years. By creating a dynamic data-rich environment, DT provides a united and reliable platform to run complicated analyses and simulations (Haag & Anderl, 2018; Tao, Zhang, Liu, & Nee, 2019). However, a data schema for dam DTs that is oriented for comprehensive evaluation purpose and a data integration/query approach that can help to automate data processing, still require further study.

Meanwhile, most evaluation methods currently adopted cannot characterise the ambiguity and randomness in the evaluation process. However, ambiguity inevitably exists in the evaluation grades (Jiang & Fan, 2008) and monitoring data collected from monitoring instruments is always discrete and random (Zhang & Li, 2019). Hence the reasoning process from measured dam behaviour to evaluation grades is full of uncertainty. In response to this situation, the cloud model, as a cognitive model that can simultaneously describe the fuzziness, randomness, and discreteness of concepts and implement the uncertain transformation between qualitative concepts and quantitative description (Li, Liu, & Gan, 2009), becoming a promising alternative. Nonetheless, the combination of necessary knowledge in the field of dam engineering with the cloud model concept to realise automated dam safety evaluation is still under investigation.

The realisation of the dynamic and proactive mode of dam safety evaluation still faces many challenges at the current stage. This paper proposes a DT-enabled solution for comprehensive evaluation of dam behaviour that contributes to tackling these challenges mentioned above and advancing the BIM provisions required to ensure dam safety during the operation and maintenance (O&M) phase of dam projects. In the presented study, a new data structure is provided based on the summarised information requirements of dam monitoring systems. Then, the related information is integrated into dam DTs by extending the industry foundation classes (IFC), which forms a fundamental basis of a reliable and robust dam evaluation program. After that, an automated evaluation framework is proposed to realise continuous dam condition monitoring. In this framework, a data classification strategy to facilitate efficient establishment of evaluation index systems is proposed. To characterise uncertainties in the evaluation process, a multi-rule cloud reasoning model is utilised to evaluate monitoring points and a multi-dimensional cloud model is leveraged to assess evaluation indices affected by multiple factors. Meanwhile, an improved Criteria Importance Though Intercriteria Correlation (CRITIC) method with dam knowledge considered is proposed to calculate weight values without human intervention.

2. Literature review

2.1. Current applications of digital twin in asset condition assessment

Nowadays, the digital twin concept has been applied to assessment analysis of building assets because of its ability to provide convenient access to comprehensive and integrated information. The information (e.g., the actual environment, applied loads and corresponding structural response) collected from Structural Health Monitoring (SHM) systems is the key ingredient of a DT (Tao et al., 2019).

Some research utilises digital models as a visualised platform for intuitive interpretation of SHM data and simple assessment of structural condition of assets. For example, Boddupalli, Sadhu, Rezazadeh Azar, and Pattyson (2019) proposed a digital representation platform based on BIM to support the visualisation of the related SHM data. Valinejadshoubi, Bagchi, and Moselhi (2019) developed a DT of a modular building to store, query, and show real-time strain monitoring data. The proposed approach supports the damage assessment of a column by setting a fixed threshold for monitored strain values. Kaewunruen et al. (2021) developed a 'digital twin + bridge risk inspection model' to achieve preventive maintenance of bridges. This platform enables the comparison of real-time monitoring data with a 'safety value' to assess structural vulnerability. There is no doubt that these studies mentioned above have highlighted the value of DT concept in asset assessment, but they only achieve informal data binding and display to perform a fast assessment by comparing monitored structural response with standard values, which does not demand high interoperability and efficient cross-referencing between integrated information.

Lu et al. (2020) proposed a DT-based anomaly detection framework to monitor the operation condition of building assets. This method enables the formal description of related information and can evaluate the mechanical condition of a centrifugal pump based on the mutual verification of its vibration monitoring data and working loads. This solution can only assess pump condition from one aspect (vibration

performance), which is not sufficient to support comprehensive evaluation. It is merely suitable for the evaluation of a single function of complex structures or building assets with a single failure mode.

However, a dam is an integrated structure which is exposed to complex natural environment. Many reasons may cause dam failure (International Commission on Large Dams (ICOLD),), 2013). Considering this, a comprehensive evaluation of dam behaviour demands the consideration of all relevant influencing factors. Also, the baselines to assess monitored variables should be deduced from the history monitoring data and condition records, which requires high-quality data integration to support complex cross-referencing. Hence a more complete and systematic data structure and an efficient data processing workflow are important to ensure data availability and interoperability. Currently, the absence of a standard data schema for the formal description of comprehensive monitoring systems and the absence of a data processing strategy for keeping data consistent are main barriers behind the continuous evaluation scenario.

2.2. Review of comprehensive evaluation of dam behaviour

The evaluation methods of dam behaviour can be classified into two categories: physical simulation analysis and data-driven methods. Physical simulation analysis mainly leverages finite element method (FEM) to calculate structural responses, with model parameters being calibrated using monitoring data (Colombo, Domaneschi, & Ghisi, 2016). Data-driven methods refer to the data fusion, modelling, and analysis of massive information to mine useful information (Su, Wen, Chen, & Tian, 2016). Among them, comprehensive evaluation of dam behaviour using multiple monitoring-related information is always a popular topic because of its significance to ensure dam safety during the O&M stage.

As shown in Figure 1, a complete evaluation index system is a multi-index and multi-layer system that includes well-organised and preprocessed monitoring data (mainly the monitored dam behaviour, such as stress, deformation, seepage, etc.). The comprehensive dam evaluation should be implemented from left to right. Hence, the reasonable and effective evaluation of each measuring point is the basis to ensure accuracy of the entire comprehensive evaluation program. After deriving the baseline of each monitoring point, the remaining challenge is to reason from a quantitative value to a qualitative index. For each monitoring point, the quantitative value is the difference between the baseline and monitored effect variable, and the qualitative index is an evaluation grade that indicates the safety status of this monitoring point, such as normal, abnormal, and so on.

Most existing research utilises the concept of fuzzy sets to evaluate safety status of monitoring points (Shu, Bao, Li, Zhang, & Wu, 2020; Wu et al., 2012). In these solutions, evaluation grades are quantified by fixed boundary conditions based on human experience, and functions of membership degree are curves with fixed parameters. Hence, the way to obtain safety status of monitoring points is deterministic. However, the truth is, ambiguity inevitably exists in the evaluation grades (Jiang & Fan, 2008) and monitoring data collected from monitoring instruments is always discrete and random because of some unpredictable factors (Zhang & Li, 2019). Thus, these existing methods cannot simultaneously characterise the ambiguity and randomness in the evaluation process of monitoring points.

Recently, several studies have adopted the cloud model concept to solve the above issue. For example, Jiang and Zhang (2013) established nine cloud models to describe the quantitative judgement matrix in the AHP index system. Jia et al. (2017) adopt cloud model to determine the basic probability assignment of the Dempster-Shafer theory. In these studies, evaluation grades are characterised using cloud models. But the uncertainties in the reasoning process from difference values to evaluation grades are ignored. Meanwhile, weight values of evaluation indices are usually determined using some subjective methods (Luo et al., 2021). There is no denying that these methods have advantages in the aspect of rich human experience, but they are both labour-intensive and time-consuming when it comes to continuous condition evaluation of whole dam structures. Hence, an automated evaluation model that can fully consider these uncertainties requires further study.

3. The IFC-based data integration for comprehensive dam evaluation

3.1. Information requirements for comprehensive evaluation of dam behaviour

Dam safety monitoring through instrumentation is the most important way to indicate dam service condition because monitoring data collected from a comprehensive monitoring system contains adequate diagnostic information which can be utilised to detect and identify abnormalities of dam behaviour. Hence, a standardised description of DSM systems is of vital importance to ensure correct classification and utilisation of various monitoring-related information.

Currently, due to insufficient understanding of the impact of multiple factors on dam safety, such as natural forces, structural mechanism, material properties, construction controls, and human behaviour, various types of dam incidents like leakage, landslide, overtopping, etc. are taking place now and then. Also, a dam is an integrated structure including the main hydraulic structure (dam body), dam foundation and abutments, blank slopes, reservoirs, and so on, incidents happened in any structure may cause dam failure. With these considerations, a systematic and complete DSM system demands comprehensive monitoring of all possible hazards in all related structures or areas.

Hence, to describe essential characteristics of monitored physical quantities, interpret dam structural behaviour, and implement monitoring-related tasks, scientific and reasonable classification of monitoring items is necessary. For example, the key monitoring items of concrete dams, prescribed in the Design Specification for Safety Monitoring in

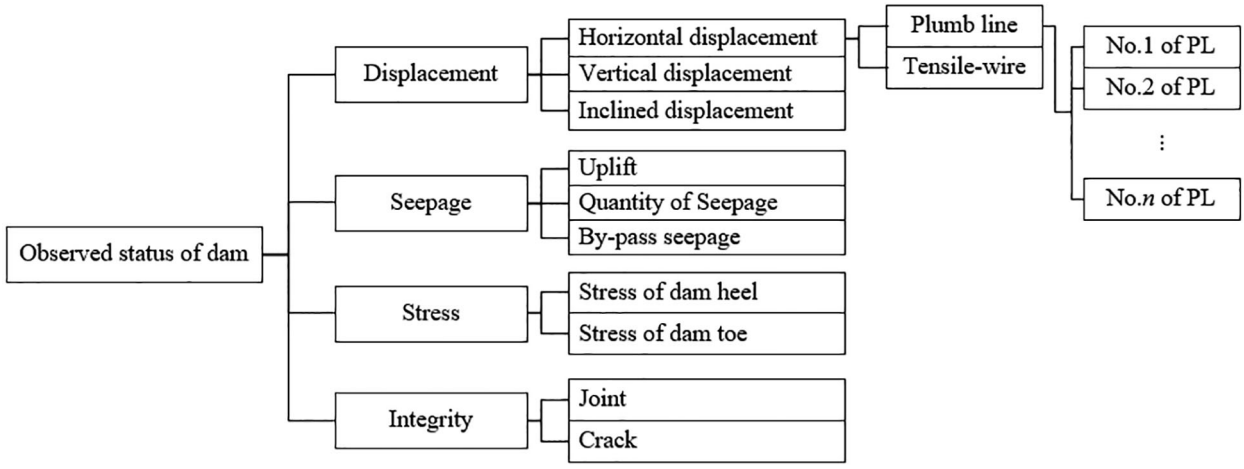


Figure 1. index system of dam health evaluation.

Water and Hydropower Projects (China SL 725-2016, 2016) are listed in Table 1. From this table, instrumental monitoring items in DSM systems are mainly divided into four categories, and each monitoring category implements specific monitoring task, which is completely different from others in this system. For instance, deformation monitoring is designed to detect shape changes of dam structures, while seepage monitoring focuses on observing the impact of groundwater movement in dams.

On the other hand, a monitoring category is composed of several monitoring items, because changes caused by external factors cannot be reflected in a single measurable variable. For instance, deformation is a comprehensive concept to describe shape changes of dams, but there is no variable that can be directly measured to characterise the extent of dam deformation. In view of this, deformation monitoring items, such as horizontal displacement, vertical displacement monitoring, and inclination, are designed to measure specific physical quantities pertaining to dam deformation from several different aspects. Using this intuitive and clear classification, instrumental monitoring-related information in the DSM system can be described in a meaningful and systematical manner.

As the most direct carrier for various monitoring data, monitoring instruments also contain a lot of useful information for preprocessing raw measured data. In current practice, monitoring instruments installed in a DSM system can be generally divided into two types: embedded monitoring instruments and external-mounted monitoring instruments (Shen & Mao, 2015). Embedded monitoring instruments refer to sensors embedded in hydraulic structures, which are mainly used to monitor physical quantities related to dam structures, such as strain, crack openings, seepage pressure, and so on. These instruments (e.g., strain gauge, osmometer, flow meter, and thermometer) are single-type monitoring instruments, that means they can be used to measure the relevant physical quantities independently without the aid of other facilities.

On the other hand, external-mounted monitoring instruments are usually installed on the surface of hydraulic structures, observation galleries, or bank slope to measure structural deformation, seepage, and environmental variables. This type of monitoring instruments, such as tensile-wire instruments, hydrostatic level instruments, and plumb-line instruments, are usually combinations of measuring sensors and some auxiliary facilities. Taking the method of tension wire alignment as an example, it is the most important and common method currently used to measure horizontal displacements of dams, and it is implemented by measuring the distances between measuring points and the datum line.

Figure 2 depicts the main components of a tensile-wire instrument: (1) endpoint devices, (2) measuring point devices, (3) measuring line, (4) protective tube, and (5) displacement sensors, among which all are auxiliary facilities except displacement sensors. Generally, auxiliary facilities only play the role of providing monitoring conditions, while these displacement sensors perform actual measurement tasks. As shown in Figure 2, a set of tensile-wire instrument is equipped with several displacement sensors, which allows to measure the horizontal displacements at multiple measuring points simultaneously. The raw measured data collected from displacement sensors which are assembled on the same set of auxiliary facilities needs to be error-corrected using the same instrument parameters and data processing method.

3.2. Data integration into DT environment

In general, DT is not only to build a 3 D digital model, but also to build an information integration platform, which includes all the life cycles of the project, and realises the storage and sharing of data. To avoid errors or data loss, a unified and standardised data schema is required as a solid foundation for complex data integration. Then, DT-enabled information management could be leveraged to simplify the process of data retrieval and utilisation. Nowadays, the IFC data schema is widely applied in the architecture, engineering, and construction (AEC) industry to describe and exchange information arise in the lifecycle of building projects (Froese et al., 1999). Hence, the use of IFC specification

Table 1. Key Monitoring items in DSM systems of concrete dams.

Monitoring Category	Monitoring Item	Monitoring Category	Monitoring Item
Deformation Monitoring	Horizontal Displacement	Stress, Strain and Temperature Monitoring	Stress
	Vertical Displacement		Strain
	Dam Inclination		Concrete Temperature
	Seam, Crack Aperture		Dam Foundation Temperature
Seepage Monitoring	Seepage	Environment Monitoring	Water Level
	Uplift Pressure		Air Temperature
	Osmotic Pressure		Reservoir Water Temperature
	Seepage Around a Dam		Sedimentation before Dam
	Water Quality		Erosion under Dam
			Ice Pressure

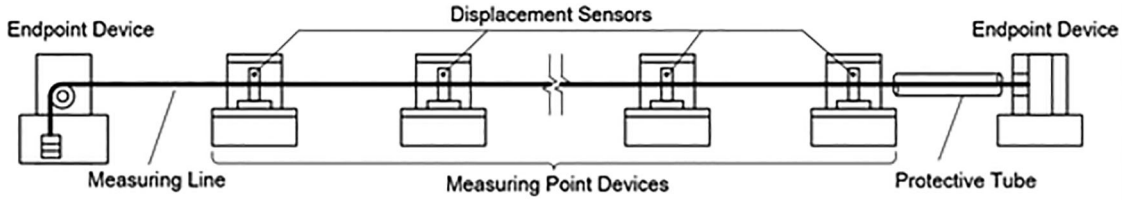


Figure 2. Structural schematic illustration of tensile-wire instrument.

for data integration and further BIM application in dam projects is promising and feasible.

As introduced in the Subsection 3.1, the information related to DSM systems is essential and fundamental to support comprehensive evaluation of dam behaviour. However, IFC schema is primarily utilised to describe conventional building information (e.g., geometry, material, cost, etc.), while the description of monitoring-related information is still in initial stage. Several extensions of the IFC schema for describing SHM systems of civil buildings have been proposed. Some research has explored the documentation of some specific components installed in SHMs. For example, Rio, Ferreira, and Martins (2013) extended the enumeration type of the entity *IfcSensor* to improve its capabilities of describing various types of structural and environmental sensors.

Other studies focus on describing detailed compositions of SHM systems and aggregation relationships between related devices from a physical and technical grouping aspect, which is specifically beneficial for facility management to ensure functionality and serviceability of monitoring devices. For instance, Theiler and Smarsly (2018) proposed a 'IFC Monitor' schema to describe semantic compositions of SHM systems and network topologies of sensor networks for the purpose of tracking changes and simplifying substitution of components installed in SHM systems. However, a systematic data schema that is oriented for the integration and efficient utilisation of dam monitoring-related information to facilitate data analysis, is still missing.

To facilitate a standardised description of monitoring-related information in DSM systems, a semantic data structure is proposed based on general regulations of dam safety monitoring, which was introduced in the Subsection 3.1. As shown in Figure 3, a DSM system is composed of several monitoring items. Monitoring items can be classified into various categories according to different monitoring tasks, such as deformation monitoring, seepage monitoring, environment monitoring and so on. Also, a monitoring item can be identified through specific measured physical quantity (e.g., horizontal displacement, vertical displacement, uplift pressure, etc.). Hence, a monitoring item can be considered as a partial system of a DSM system that can measure, collect, and send effect values of the same physical quantity.

Some common categories and corresponding monitoring items in the context of dam safety monitoring for concrete dams are introduced in Table 1. For earth-rockfill dams and other hydraulic structures, these monitoring categories are the same as those of concrete dams, but some different monitoring items for measuring special physical quantities may be included. The set-up of monitoring items can facilitate the implementation of dam safety monitoring and support efficient data collection in accordance with monitoring index systems for comprehensive dam evaluation.

Furthermore, a monitoring item is a sensory network designed to measure, collect, and send the same variable attribute. Each monitoring item comprises sensors, sensor assemblies, and programmable logic controllers (PLCs). Sensor assemblies are defined for representing external-mounted monitoring instruments, while sensors are used to characterise embedded monitoring instruments. Generally, for different monitoring items, sensors or sensor assembly are selected according to the attribute of the measured physical quantity. However, at least one sensor or sensor assembly is required in a monitoring item to detect dam behaviour, also at least one PLC is attached to receive, decode, and then send the measured data. Moreover, considering the characteristic of external-mounted monitoring instruments, monitoring facilities and sensors are necessary components of a sensor assembly. Here, the monitoring facility can be regarded as a combination of all the auxiliary facilities in an external-mounted monitoring instrument. Finally, semantic references from sensors to structural models of dams are defined because the positions of measuring sensors in dam structures are essential to ensure the meaningfulness of the measured data.

Based on the proposed data structure, some new entities, enumeration types, and properties need to be extended for the integration of monitoring-related information into DT environment. In this study, the extension, including three entities, is proposed based on the investigation of the current

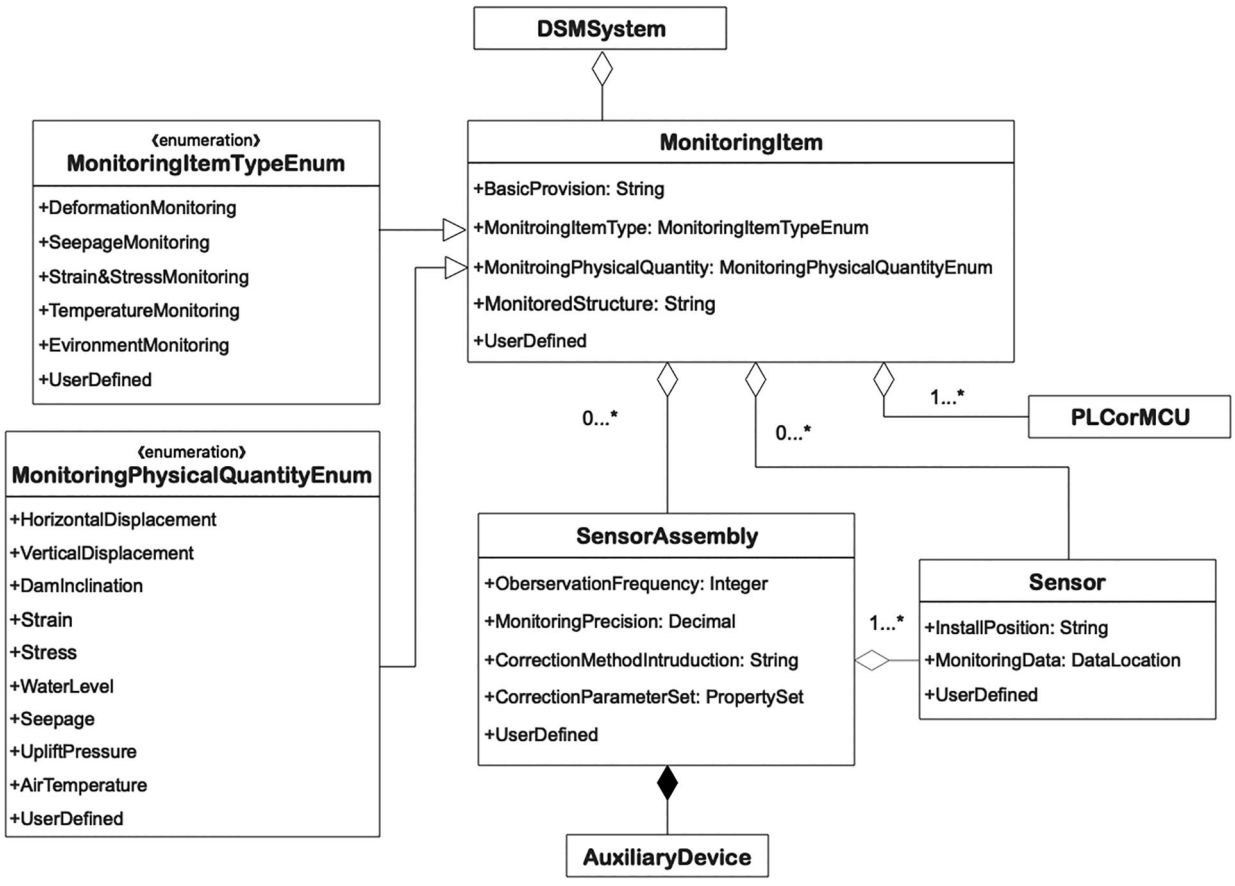


Figure 3. A semantic data structure for describing a DSM system.

IFC schema (IFC 4 Addendum2) and information requirements for modelling DSM systems. Figure 4 shows a diagram of the selected IFC4 entities. The newly proposed and extended entities are shaded in dark grey, and the entities predefined in the current IFC schema, which are required to model DSM systems, are shaded in light grey. In this extension, the *IfcSensorAssembly* entity, which is designed to describe sensor assemblies, is a subtype of the *IfcElementAssembly* entity. A monitoring item is described by *IfcMonitoringItem*, which is a subtype of the entity *IfcSystem*. A dam safety monitoring system (*IfcDSMSystem*) is an organised combination of related partial systems, which is composed to provide monitoring service for dams. The enumeration types and detailed properties of each newly extended entity can be designed directly according to the proposed data structure. For monitoring instruments or their properties that do not have corresponding items in the current IFC schema, the extension method proposed by Rio et al. (2013) is used as a reference. For example, the location of monitoring data is saved as the value of the 'MonitoringData' property of the corresponding monitoring instrument.

4. Cloud model-based comprehensive evaluation of dam safety status

4.1. Construction of evaluation index system

Data collection and processing for comprehensive evaluation of dam behaviour is both labour-intensive and time-consuming. Till now, all the related information is well-defined and integrated into dam DTs based on the extended IFC schema, the time required for information query, as well as the possibility of errors are reduced (Ding et al., 2009). However, it is still insufficient to fully support the evaluation of dam behaviour because diagnostic information needs to be classified according to the used evaluation index system. As shown in Figure 1, the result of dam safety evaluation does not depend on the observed characteristic from a single monitoring item or a single observation point, but on the comprehensive safety condition derived from all evaluation indices (with different weight values). Usually the first-level index (highest level) is the overall safety status of the dam being evaluated, other evaluation indices shown in the second level are determined by possible failure modes of this dam. To clarity the impact of each failure mode on dam safety, dam engineers need to categorise various monitoring points according to these indices.

The classification process in the traditional mode is performed as follows. First, monitoring data is collected from databases, as well as the corresponding sensor number, monitoring type, monitoring physical quantity, etc. Secondly, position information of this sensor is obtained by checking the monitoring instrument layout. Thirdly, based on above information, the index to which this monitoring point belongs can be derived. During this process, human intervention is necessary to collect and judge the related information (e.g., monitoring type, position, etc.) of each

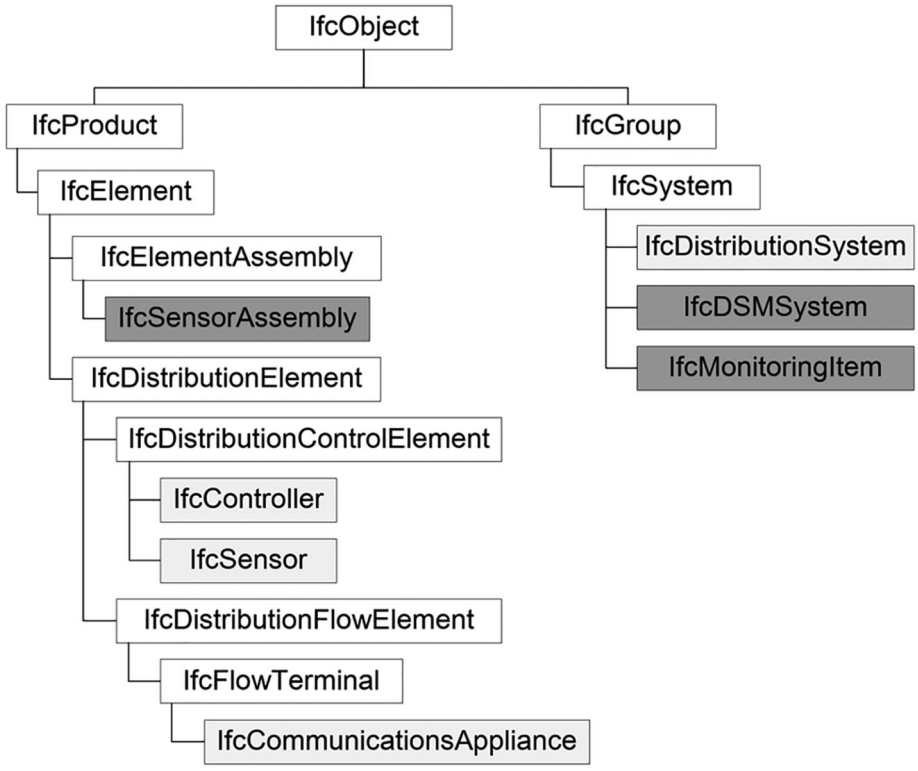


Figure 4. Newly proposed and predefined IFC entities required to model a DSM system.

monitoring instrument, which is labour-intensive and error prone.

To improve the efficiency of dam behaviour evaluation, a new strategy is proposed to facilitate the automated data classification. In current practice, failure modes of dams can be described from two aspects: (1) specific reasons of dam incidents, or (2) specific structures of dam incidents. The sample evaluation index system shown in Figure 1 is organised following the first way. That is, the second-level indices represent four possible reasons of dam incidents, and the third-level indices are the corresponding physical quantities. Some other evaluation index systems (e.g., (Shu et al., 2020)), which are constructed following the second way, are also of practical significance in dam engineering. For example, their second-level indices represent some possible structures (e.g., dam body, foundation, abutment, etc.) of dam incidents, and the third-level indices are the corresponding physical quantities. The common point of these two construction methods is that third-level indices are always monitoring physical quantities. Meanwhile, as introduced in Section 3, monitoring items to measure the same physical quantities are defined in a standard manner. Therefore, a monitoring item can be regarded as an abstract object for data query and classification.

A new classification process is introduced as follows. First, according to the second-level indices, the monitoring category or the monitored structure is chosen as the query condition to determine the corresponding monitoring items. Then, for each monitoring item, all the related monitoring instruments or sensors are obtained by relational entities. Finally, the monitoring data is extracted for assessment. This method is workable for the following reasons. First, monitoring items can be formally described using the extended IFC entity *IfcMonitoringItem*. In the traditional method of monitoring-related information management, a monitoring item, as a collection of monitoring instruments or sensors that measure the same physical quantity, is an abstract concept which is not clearly defined. Hence it cannot be queried conveniently and correctly. Secondly, all the information required to support this classification process is integrated into the dam DT. The query conditions (e.g., monitoring category and monitoring structure) are well-defined as the properties of the corresponding entities, thereby ensuring the availability of various required information.

In the proposed method, the monitoring items are classified first, then the corresponding monitoring instruments are queried, and finally the monitoring data is extracted. The advantages of this method are shown in the following aspects. First, the classification objects are changed from thousands of monitoring instruments to relatively few monitoring items. Hence the efficiency of classification is improved as the number of judgements is reduced. Secondly, this method is proposed based on the well-defined data structure of monitoring-related information. The monitoring data is saved as a property of monitoring instruments, which is of vital importance to fully support the automated data processing and can be widely applied in dam engineering.

4.2. Measurement of evaluation index

The assessment of evaluation indices is essentially the analysis of the related monitoring points. The safety status of a

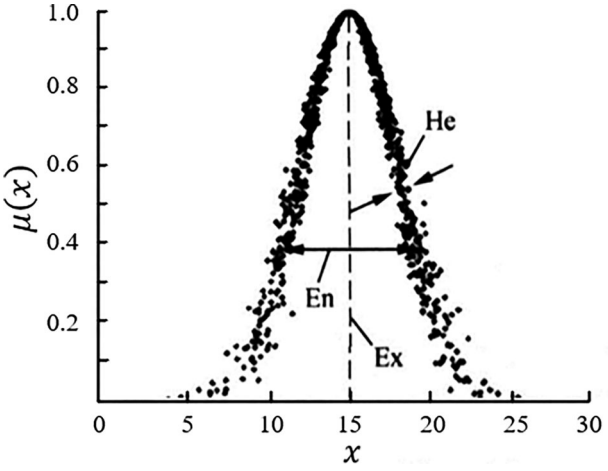


Figure 5. A one-dimensional cloud model.

measuring point can be judged by comparing the monitoring data with its corresponding evaluation criterion or baseline. These evaluation criteria change with the loads on the dam and the external environment. In dam engineering, evaluation criteria for monitoring points are usually predicted values, which can be derived from simulation calculation of FEM or analysis results of monitoring data-based models. The FEM requires timely updating of model parameters to calculate accurate prediction values of structural response. Currently, this part of work highly relies on human intervention. Hence, for continuous evaluation, the FEM is not efficient enough to provide updated evaluation criteria. The data-based models of monitoring points can be established based on the real relationships between the loads or environment (e.g., water level, temperature, time) and structural response (e.g., deformation, seepage, crack openings, etc.).

The widely used method to establish data-based models is statistical analysis. These statistical models of monitoring points, also known as quantitative interpretation models, have the advantages of clear meaning and easy implementation. For example, the hydrostatic-seasonal-time (HST) model is the most famous statistical model for interpreting horizontal displacements of dams (Salazar, Morán, Toledo, & Oñate, 2017). The HST model is written as:

$$\begin{aligned}\delta(h,s,t) &= a_0 + f(h) + f(s) + f(t)\\ \delta &= a_0 + \sum_{i=1}^{3}\left[a_i\left(h^i - h_0^i\right)\right] + \sum_{i=1}^{2}[b_{1i}(\sin is - \sin is_0)\\ &\quad + b_{2i}(\cos is - \cos is_0)] + c_1(t - t_0) + c_2(lnt - lnt_0)\end{aligned} \tag{1}$$

where δ is the horizontal displacement of dams, h is the difference of water level between upstream and downstream, s is the season varying between 0 and 2π from January 1 to December 31 according to $s = (j/365)2\pi$ with j being the number of days since January 1, and t is the time from the monitoring started to the data was collected. The model coefficients can be deduced by multiple linear regression (MLR) method. The statistical models for other structural response (e.g., crack openings (Dai et al., 2019), seepage (Wei, Gu, Li, Xiong, & Xu, 2018) of dams have also been proposed. For each monitoring point, after the statistical model is in place, the predicted dam response under certain conditions can be deduced.

As shown in Equation (1), the information cross-referencing behind the establishment of these statistical models is complex and time-consuming. Fortunately, the IFC-enabled data integration facilitates the standardised description of various information and enables more efficient data access, thus data availability is enhanced. Meanwhile, the cross-referencing is performed based on the well-defined and object-oriented data integration method, which highly reduce the time of information extraction and the probability of errors.

4.3. Automated comprehensive evaluation framework for dam safety

4.3.1. Multi-dimensional normal cloud model

In the theory of cloud model (Li et al., 2009), three basic concepts are defined to describe digital characteristics of a qualitative concept, and each cloud drop $(x_i,\ \mu_i)$ represents a concrete realisation of this qualitative concept. The expectation value E_x describes the distribution expectation of cloud drops in the domain space of a qualitative concept. The entropy E_n represents the degree of dispersion of cloud drops that can describe this qualitative concept. The hyper-entropy H_e is determined by the randomness and ambiguity of entropy E_n. Assume that U is a quantitative domain expressed with precise numerical values, C is the qualitative concept of U, and the three digital characteristics of C is (Ex, En, He). If $x \in U$, and it is a random instantiation of concept C. Also, if $x \sim N\left(E_x, (En')^2\right)$, $En' \sim N(E_n, He^2)$, and the membership degree $\mu(x)$ of x to C satisfies:

$$\mu(x) = e^{-\frac{(x-E_x)^2}{2(En')^2}} \tag{2}$$

Then, the distribution of x in the domain U is defined as a one-dimensional normal cloud (Figure 5), and $(x, \mu(x))$ is one of the drop clouds.

The three digital characteristics of qualitative concepts restricted by m-dimensional factors are defined as $(Ex_1, En_1,\ He_1)$, $(Ex_2, En_2,\ He_2)$, …, $(Ex_m, En_m,\ He_m)$. The construction of a m-dimensional normal cloud model can be realised by a forward m-dimensional cloud generator (CG) (as shown in Figure 6). The main steps are:

1. Generate a normal random number $X(x_1, x_2, \ldots, x_m)$ with $Ex(Ex_1, Ex_2, \ldots, Ex_m)$ as the expectation and $En(En_1, En_2, \ldots, En_m)$ as the standard deviation.
2. Generate a normal random number $En'(En'_1, En'_2, \ldots, En'_m)$ with $En(En_1, En_2, \ldots, En_m)$ as the expectation and $Hn(He_1, He_2, \ldots, He_m)$ as the standard deviation.
3. Calculate the corresponding membership degree: $\mu\left(X(x_1, x_2, \ldots, x_m)\right) = e^{-\sum_{i=1}^{m}\frac{(x_i-Ex_i)^2}{2(En'_i)^2}}$, $i = 1, 2, \ldots, m$. Then

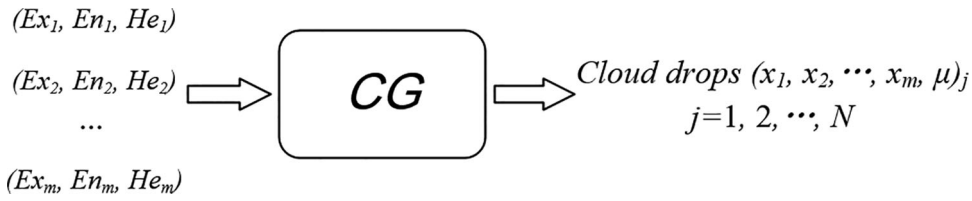


Figure 6. Schematic diagram of a m-dimensional forward cloud generator.

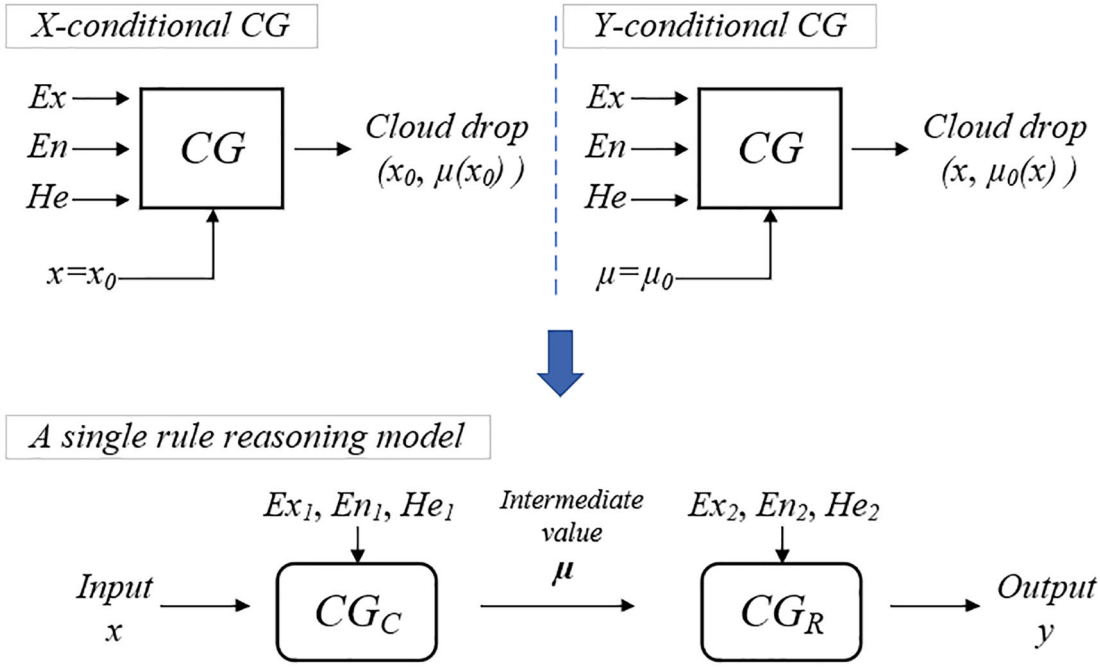


Figure 7. A single rule cloud reasoning model.

$(X(x_1, x_2, ..., x_m),\ \mu(X(x_1, x_2, ..., x_m)))$ is one of cloud drops of this generated multidimensional cloud model.

4. Repeat steps (1) ∼ (3) until N cloud drops are generated.

In steps (1) ∼ (3), the cloud drops generated by CG are neither unique nor definite because the X and the En' are random numbers. This is the fundamental reason why the cloud model can consider randomness and fuzziness at the same time.

4.3.2. Multi-rule cloud reasoning model

In common practice, a typical single-rule reasoning process can be expressed as '*if condition, then result*', and a multi-rule reasoning process is '*if condition 1, then result 1; if condition 2, then result 2; … if condition n, then result n.*' In the cloud model, the single-rule reasoning process can be realised by the combination of a X-conditional CG and a Y-conditional CG (Figure 7).

- X-conditional CG: Input: the given digital characteristic (Ex_1, En_1, He_1) of a *condition* concept, and the given value x to be inferred; Output: the corresponding membership degree μ.
- Y-conditional CG: Input: the given digital characteristic (Ex_2, En_2, He_2) of a *result* concept, and the calculated membership degree μ; Output: the y value that satisfies the membership degree μ.

A multi-rule cloud reasoning model is an extension of the single-rule cloud reasoning model, the main steps of a n-rules cloud reasoning model are:

1. For every rule $(i = 1, 2, ..., n)$ and the digital characteristics (Ex_i, En_i, He_i) of the corresponding *condition* i, generate a random number En'_i with the En_i as the expectation and the He_i as the standard deviation.
2. Calculate the degree of membership μ_i of the input value x to each rule using:

$$\mu_i = e^{-\frac{(x-E_{xi})^2}{2En'^2_i}} \tag{3}$$

3. Select the largest degree of membership: $\mu = max(\mu_1, \mu_2, ..., \mu_n) = \mu_m$.
4. According to the digital characteristics (Ex'_m, En'_m, He'_m) of the *result* of *rule* m, generate a normal number En''_m with the En'_m as the expectation and the He'_m as the standard deviation.
5. Solve the y value according to the following equation inversely.

$$\mu_m = e^{-\frac{(y-E'_{xm})^2}{2En''_m{}^2}} \quad (4)$$

6. Repeat steps (1) ~ (5) k times, the result is the mean of these y values.

4.3.3. Improved criteria importance though intercriteria correlation (CRITIC) method

Usually, a monitoring item is composed of numerous monitoring points which play large or small roles in comprehensive monitoring. Hence, to obtain the safety status of a monitoring item, the weight values of all related monitoring points are necessary. In this study, the multi-correlation coefficients of their statistical models are utilised as a dam knowledge to improve the classic Criteria Importance Though Intercriteria Correlation (CRITIC) method (Diakoulaki, Mavrotas, & Papayannakis, 1995). The multi-correlation coefficient which represents the fitting degree of each monitoring point is an important reference for the accuracy of the predicted dam response. Meanwhile, for a monitoring point, the larger multi-correlation coefficient means the bigger influence of applied loads on dam response, that is, containing more diagnostic information (Su et al., 2016).

The process to calculate weight values of these related monitoring points is introduced below. Assume that, for a monitoring item, the number of all related monitoring points is m, and monitoring data at n time points are used to evaluate. The main steps are as follows:

1. Suppose the original monitoring data matrix is $Y = [y_{kt}]_{m\times n}$. Utilising zero-mean normalisation (Z-sore) method, the normalised matrix expressed as $Z = [z_{kt}]_{m\times n}$ can be calculated by:

$$z_{kt} = \frac{y_{kt} - \overline{y_k}}{s_k} \quad (k = 1, 2, \ldots, m; t = 1, 2, \ldots, n) \quad (5)$$

where, $\overline{y_k} = \frac{1}{n}\sum_{t=1}^{n} y_{kt}$ is the mean monitoring data of each measuring point. $s_k = \sqrt{\frac{1}{n-1}\sum_{t=1}^{n}(y_{kt} - \overline{y_k})^2}$ is the standard deviation of monitoring data of each measuring point.

2. The coefficient of variation δ_k can be calculated by:

$$\delta_k = s_k / \overline{y_k} \quad (k = 1, 2, \ldots, m) \quad (6)$$

3. The correlation matrix $R = [\gamma_{kt}]_{m\times n}$ of the normalised matrix Z is calculated.
4. The quantitative coefficient η_k of the degree of independence of each monitoring point can be obtained by:

$$\eta_k = \sum_{t=1}^{n}(1 - \gamma_{kt}) \quad (k = 1, 2, \ldots, m) \quad (7)$$

5. The comprehensive information C_k of each measuring point is determined by Equation (8). In this step, the multi-correlation coefficient r_k of statistical model of each measuring point is considered:

$$C_k = \delta_k \eta_k r_k \quad (k = 1, 2, \ldots, m) \quad (8)$$

6. Finally, the weight value of each monitoring point can be calculated by:

$$\omega_k = \frac{C_k}{\sum_{k=1}^{m} C_k} \quad (k = 1, 2, \ldots, m) \quad (9)$$

4.3.4. The proposed evaluation framework

The proposed framework for comprehensive evaluation of dam safety is shown in Figure 8. In this solution, data availability is provided and enhanced by DT-based data integration. The whole evaluation process is divided into three parts. First, the first-level assessment is to evaluate the safety status of each monitoring item, that is, the related monitoring points. For each monitoring point, the raw measuring data is corrected using correction method and the evaluation criterion or baseline used to indicate the standard dam response is deduced from the corresponding statistical model. The input of a statistical model is the consistent long-term monitoring data, more specifically, the historical structural response (e.g., displacement, stress and strain, joint, etc.) and corresponding working conditions (e.g., water level, temperature, time effect, etc.). The outputs are regression coefficients of a quantitative interpretation function which represents the real relationship between structural response and the loads or environment, and a multi-correlation coefficient which represents the fitting degree of this statistical model.

Using this model, dam response under certain conditions can be predicted. According to the difference between the measured and predicted structural behaviour, the health status of a measuring point can be inferred by a multi-rule cloud reasoning model. The evaluation rules are necessary to establish a multi-rule cloud reasoning model. For example, the qualitative rules could be '*if the difference is very small, then the health status is normal*' or '*if the difference is a little large, then the health status is little abnormal*'. The three digital characteristics (Ex, En, He) of conditions and results in rules can be determined by specific dams. The input of the multi-rule cloud reasoning model is the difference between the measured and predicted structural response, the output is an effect score which represents the safety status of the monitoring point. The inputs of the improved CRITIC method are the effect scores and multi-correlation coefficients of all the related monitoring points in a monitoring item. The outputs are corresponding weight values. Then, the safety status of the monitoring item is the weighted average of all effect scores.

Secondly, the second-level evaluation indices are assessed. In this step, safety status of related monitoring items can be regarded as influencing factors. The weight values of these factors are calculated by the entropy weight method (EWM). Then, a multidimensional cloud evaluation model is established, and the evaluation grades for describing dam safety status which are expressed as three digital characteristics (Ex, En, He) are determined by specific dams in advance. The inputs are safety status of all related monitoring items and corresponding weight values. The output is a matrix of membership degree of the safety status for each evaluation grade. Thirdly, the third-level assessment is to evaluate the safety status of a whole dam. The weight values of these

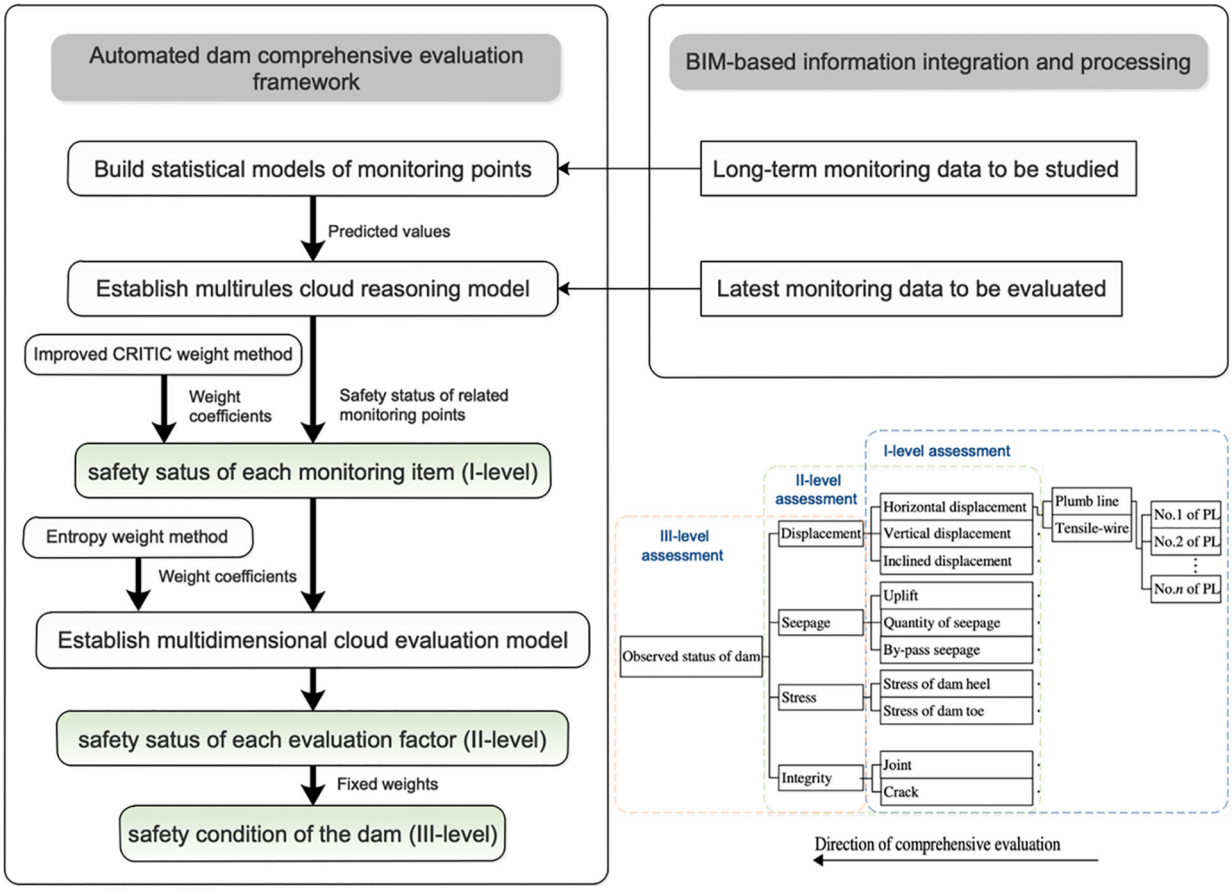


Figure 8. The framework for comprehensive evaluation of dam safety.

evaluation factors, which can be determined by dam engineer with rich experience, are fixed values in a continuous evaluation scenario. The output is a weighted average of matrices of membership degree of all evaluation factors. Then, according to the maximum membership principle (Civanlar & Trussell, 1986), the safety status of the whole dam can be inferred.

5. Case study

In this section, the extended IFC schema is used to describe a prototype DSM system installed in the Jiangya dam in China to test and verify its modelling capability. Then, a continuous safety evaluation is performed based on the proposed framework and dynamic information provided by the dam DT. The objectives of this case study are to demonstrate how the designed data structure and proposed evaluation framework can contribute to standardised integration of monitoring-related information, continuous dam safety evaluation, and potential problem detection during the O&M stage of dams.

5.1. Ifc-based data integration into the dam DT

For clearer illustration, only a part of the prototype DSM system is shown in Figure 9. The main hydraulic structure of this dam project is a gravity dam with a height of 131 m and a length of 368 m. As shown in Figure 9, there are three observation galleries inside the dam and each of them is equipped with monitoring instruments. In the first observation gallery, a set of tensile-wire instrument and a set of hydrostatic level instrument are installed to measure the horizontal and vertical displacements of the dam body at the elevation of 200.0 m, respectively. $LS_{1\text{-}1}$~$LS_{1\text{-}12}$ represent displacement sensors included in the tensile-wire instrument, and $EX_{1\text{-}1}$~$EX_{1\text{-}10}$ represent displacement sensors contained in the hydrostatic level instrument. Similarly, the second observation gallery at the elevation of 160.0 m undertakes the same monitoring tasks. Meanwhile, twelve uplift pressure gauges are buried in the dam concrete below the third observation gallery to measure the uplift pressure at the dam foundation. All these monitoring instruments are installed in a well-designed wireless sensor network, which is a premise for integrated sensing and real-time information extraction.

The DSM system involves three monitoring items. To implement different monitoring tasks, both external-mounted and embedded monitoring instruments are used, i.e., the hydrostatic level instruments, tensile-wire instruments, and uplift pressure gauges. Other components of the DSM system, e.g., the PLCs, computer center, and communication network, are not marked on Figure 9 because they can be modelled by the current IFC schema and are not the focus of this study. Then, using the BIM toolkit xBIM (eXtensible Building Information Modelling) (Lockley,

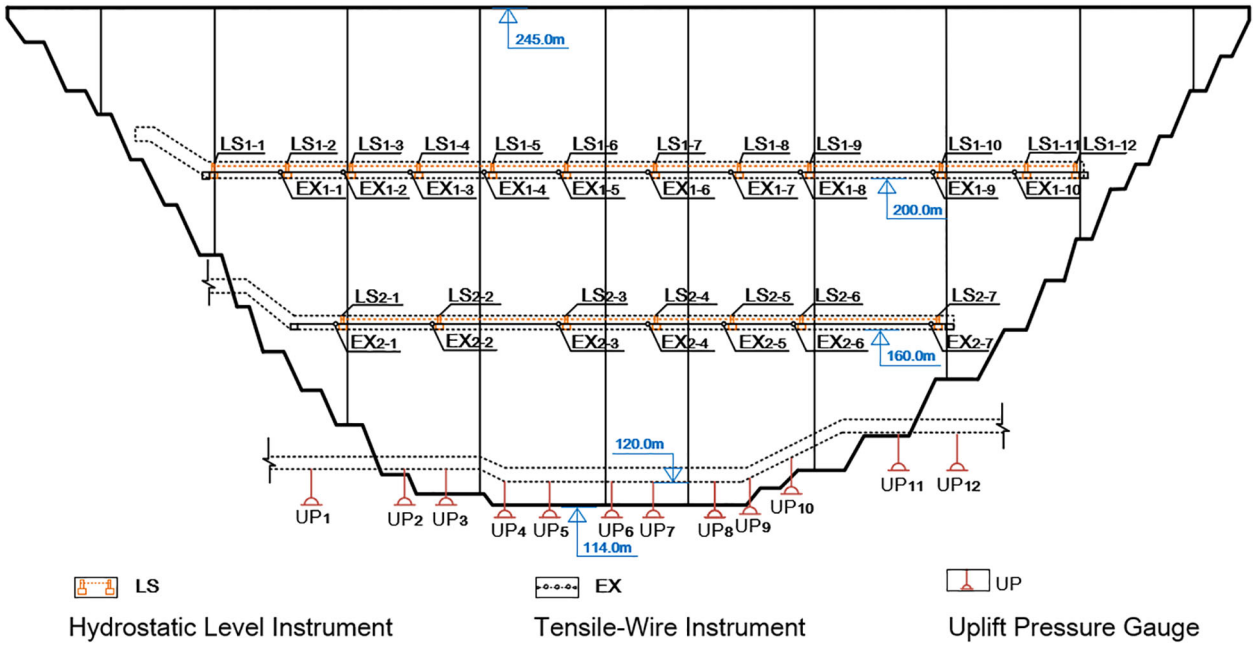


Figure 9. Monitoring layout of the selected DSM system.

Benghi, & Černý, 2017), the DSM-related information is described based on the extended IFC schema. In Figure 10, the composition relationships of the compliant IFC model are represented. The detailed properties of instances have been omitted in the figure but are briefly introduced in the following.

As shown in the upper part of Figure 10, the prototype DSM system (#1) is composed of three monitoring items (#2, #3, #4), a communication network (#66), and a computer center (#67). The monitoring item type of the first and second monitoring item (#2 and #3) is deformation monitoring, while the third monitoring item (#3) is seepage monitoring. Meanwhile, the monitoring physical quantities of monitoring items #2, #3, and #4 are horizontal displacement, vertical displacement, and uplift pressure, respectively. The instance (#91) of the relationship entity *IfcRelAssignsToGroup* is used to describe the grouping relationship between the DSM system and its components, i.e., three monitoring items (#2, #3, #4), a communication network (#66), and a computer center (#67).

In addition, the monitoring item (#2) is composed of two sensor assemblies (#5, #6), typed by wire alignment instrument, and a PLC (#22) described by *IfcController*. The monitoring item (#3) is composed of two sensor assemblies (#7, #8) typed by hydrostatic levelling instrument, and a PLC (#23). The monitoring item (#4) includes twelve sensors (#9∼#20) to detect uplift pressure on the base of the dam, and a PLC (#21). Also, the grouping relationship between monitoring items (#2, #3, and #4) and related objects (sensor assemblies or sensors, PLCs) are described by the instances (#92, #93, and #94) of the *IfcRelAssignsToGroup* entity, respectively.

As shown in the lower part of Figure 10, the sensor assembly (#6) is taken as an example to illustrate the composition structure of sensor assemblies. Here, sensor assembly #6 is composed of seven components, including six displacement sensors (#27∼#32) and a measuring device (#26). An instance (#99) of the relationship entity *IfcRelAggregates* is used to gather related elements to the sensor assembly. Furthermore, the relationship between sensors and the dam structure (#70) being monitored is defined using the instance (#100) of the entity *IfcRelContainedInSpatialStructure*.

5.2. Comprehensive dam safety evaluation

In this case, the evaluation index system is organised following Figure 1. The instance #2 of entity *IfcMonitoringItem* is utilised to introduce the process of data collection and utilisation. According to the second-level indices of the evaluation index system, the 'MonitoringItemType' of entity *IfcMonitoringItem* is used as query condition to classify monitoring items. For example, the monitoring item type of the instance #2 is deformation monitoring, hence, its position in the third-level indices can be determined. Then, sensors or sensor assemblies attached to this monitoring item are obtained based on its composition relationships. More specifically, two sensor assemblies (#5, #6) belong to the monitoring item #2.

For each sensor assembly, to obtain the real effect quantity of all related sensors, correction method and parameters, as well as installation position of measuring points are used to correct the raw measured data first. Then, the statistical model of each monitoring point is established based on its historical monitoring data and cross-referencing of other information (e.g., water level and environment monitoring data, monitoring date, installation position, etc.). Note that monitoring points related to the same monitoring item share the same form of statistical model, but the model coefficients are different. After that, the predicted dam response on each monitoring point is calculated.

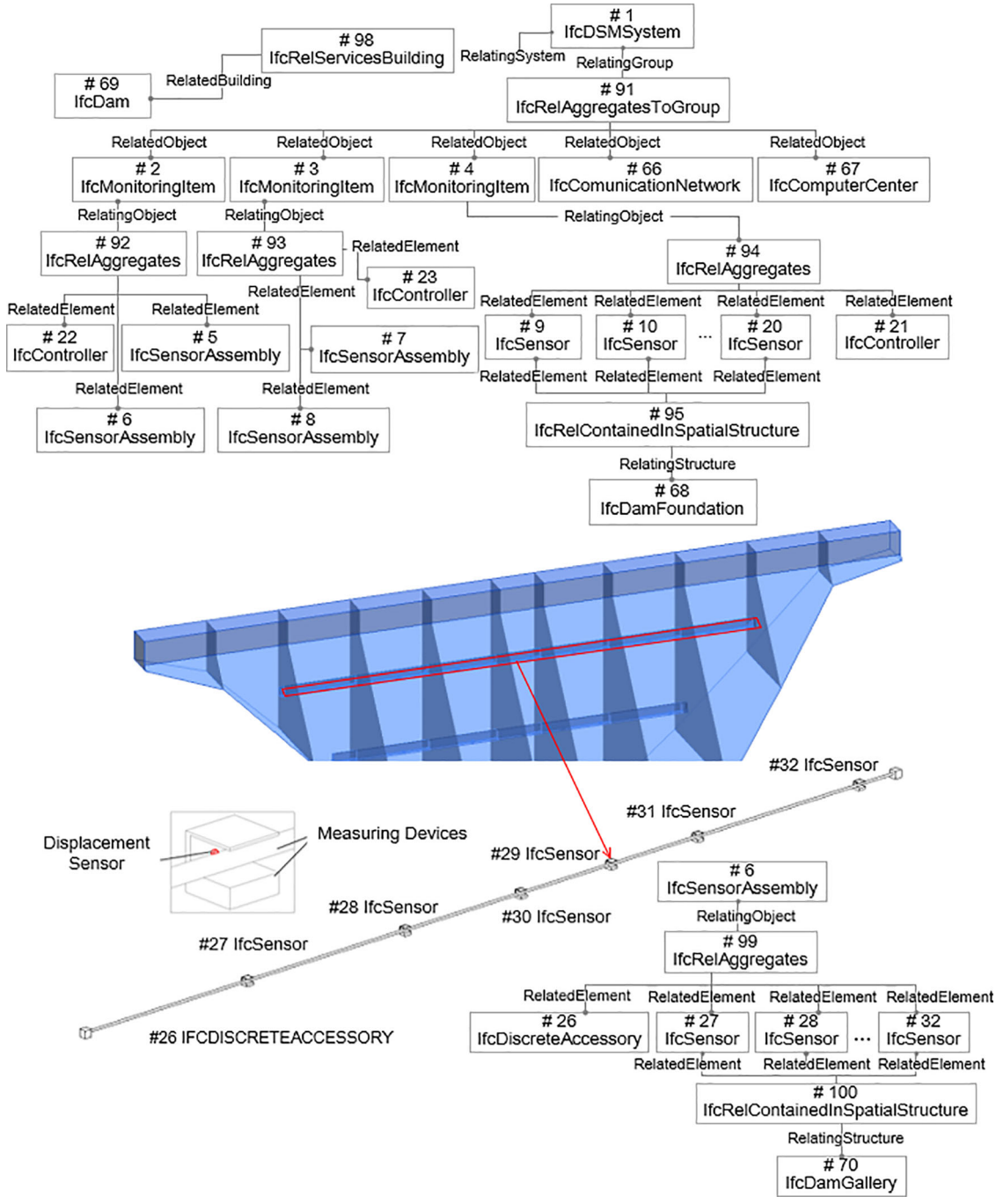


Figure 10. The composition relationships of the DSM system modelled by the IFC schema.

For example, the statistical model of the displacement sensor (instance #29) is Equation (1). Based on the cross-referencing of monitoring information from January 2010 to October 2018, the model coefficients are calculated, as shown in Table 2. Meanwhile, the multi-correlation coefficient of this model is 0.927, which will be used to improve the weight value of this monitoring point. Using this model, the normal horizontal displacements from September 1st to September 7th, 2019 at the monitoring point #29 are predicted as evaluation criteria. The real monitoring data and predicted values are shown in Table 5. In this case study, to eliminate measuring errors, monitoring data of several consecutive days is used to assess dam safety status.

Table 2. Model coefficients of the statistical model of a displacement sensor (#29 instance).

a_0	a_1	a_2	a_3	b_{11}
−5.10E-01	−9.53E-01	1.58E-02	−7.48E-05	1.03E + 00
b_{21}	b_{12}	b_{22}	c_1	c_2
1.27E + 00	−2.92E-01	1.41E-01	3.26E-02	0.00E + 00

Then, to assess each monitoring point, the next step is to establish a multi-rule cloud reasoning model. The evaluation rules are shown in Table 3, where y is the monitoring data to be evaluated, $\hat{y}$ are the predicted values, and S is the standard deviation of these differences. Then, for each rule,

Table 3. Evaluation rules of the established a multi-rule cloud reasoning model.

Evaluation Rules
Rule1: if $\|y-\hat{y}\| < S$, then the score which represents the safety status is between [0,0.2)
Rule2: if $S \le \|y-\hat{y}\| < 2S$, then the score which represents the safety status is between [0.2,0.4)
Rule3: if $2S \le \|y-\hat{y}\| < 3S$, then the score which represents the safety status is between [0.4,0.6)
Rule4: if $3S \le \|y-\hat{y}\| < 4S$, then the score which represents the safety status is between [0.6,0.8)
Rule5: if $\|y-\hat{y}\| > 4S$, then the score which represents the safety status is between [0.8,1.0]

Table 4. The digital characteristics of concepts in the reasoning model of #29 instance.

$(Ex, En, He)_C$	$(Ex, En, He)_R$
(0.0569, 0.0189, 0.01)	(0.1, 0.0333, 0.02)
(0.1707, 0.0189, 0.01)	(0.3, 0.0333, 0.02)
(0.2844, 0.0189, 0.01)	(0.5, 0.0333, 0.02)
(0.3982, 0.0189, 0.01)	(0.7, 0.0333, 0.02)
(0.4551, 0.0189, 0.01)	(0.9, 0.0333, 0.02)

Table 5. The scores representing the safety status of #29 instance.

Monitoring date	Real monitoring data (mm)	Predicted value (mm)	Difference (mm)	Score
2019.9.1	−3.60426	−3.45	−0.15426	0.3405
2019.9.2	−3.59680	−3.39	−0.20680	0.4111
2019.9.3	−4.05266	−4.03	−0.02266	0.1951
2019.9.4	−4.09790	−4.13	0.03210	0.1623
2019.9.5	−4.26877	−4.32	0.05123	0.1135
2019.9.6	−4.34694	−4.46	0.11306	0.2993
2019.9.7	−3.60426	−3.45	0.15426	0.3232

Table 6. Evaluation results of dam deformation index.

Evaluation Index	μ_{N1}	μ_{N2}	μ_{N3}	μ_{N4}	μ_{N5}
Dam Deformation	0.0000	0.4011	0.1838	0.0897	0.0548

Table 7. Evaluation results of comprehensive dam safety in September 2019.

Evaluation Index	μ_{N1}	μ_{N2}	μ_{N3}	μ_{N4}	μ_{N5}
Dam Safety	0.0013	0.4591	0.1473	0.0701	0.0434

three digital characteristics of the condition and result are calculated to blur the fixed boundary, thereby the randomness and discreteness of concepts can be considered. If a concept has an upper and a lower boundary, like $[V_{min}, V_{max}]$. Then the expectation Ex can be determined by $E_x = (V_{min} + V_{max})/2$, the entropy E_n can be determined by $E_n = (V_{max} - V_{min})/6$, and the hyper-entropy H_e can be regarded as a fixed value. Else if a concept has only a single boundary, like: $[-\infty, V_{max}]$ or $[V_{min}, +\infty]$, then the expectation Ex can be determined by $E_x = V_{min}$ or V_{max}, and the entropy E_n and the hyper-entropy H_e can be regarded as fixed values.

The digital characteristics of the reasoning model of the instance #29 is listed in Table 4 as an example. In Table 4, $(Ex, En, He)_C$ represents the digital characteristics of condition concept in rules, and $(Ex, En, He)_R$ is the result concept. After evaluation rules are set up, the scores which represent the safety status of the instance #29 are deduced based on the real monitoring data and predicted values, the results are shown in Table 5. Following the steps above, all monitoring points belong to the monitoring item #2 are assessed. Then the improved CRITIC method is used to calculate weight values, and the weighted scores of these points represent the safety status of the monitoring item #2.

Table 8. Evaluation results of comprehensive dam safety from January 2019 to September 2019.

Date	μ_{N1}	μ_{N2}	μ_{N3}	μ_{N4}	μ_{N5}
Jan	0.0000	0.4333	0.1427	0.0489	0.0595
Feb	0.0000	0.4600	0.1454	0.0794	0.0697
Mar	0.0000	0.4714	0.1740	0.0616	0.0662
Apr	0.0016	0.4316	0.1377	0.0637	0.0424
May	0.0014	0.4217	0.1578	0.0615	0.0561
Jun	0.0015	0.4515	0.1911	0.0651	0.0584
Jul	0.0013	0.4215	0.1678	0.0573	0.0368
Aug	0.0014	0.4342	0.1265	0.0730	0.0218
Sep	0.0013	0.4591	0.1473	0.0701	0.0434

After all monitoring items are evaluated, the second-level indices can be assessed. For each evaluation index in the second level, the multi-dimensional cloud evaluation model is established as following. First, based on the human experience and common grading standards, the safety status of each index is represented using five grades. The evaluation grade set is described as $N_l = \{N_1, N_2, N_3, N_4, N_5\} = \{$Normal, Basically Normal, Little Abnormal, Abnormal, Dangerous$\}$. Also, according to the range and principle (The smaller the value is, the better the safety is) of normalised value of evaluation indices, corresponding quantitative interval values are set as $V = \{[0, 0.2), [0.2, 0.4), [0.4, 0.6), [0.6, 0.8), [0.8, 1.0]\}$. Secondly, the digital characteristics of evaluation grades are calculated using the method mentioned above. Thirdly, the weight values of all related monitoring items are calculated using the entropy weight method. Then, using the cloud evaluation model, each evaluation index can be assessed. For example,

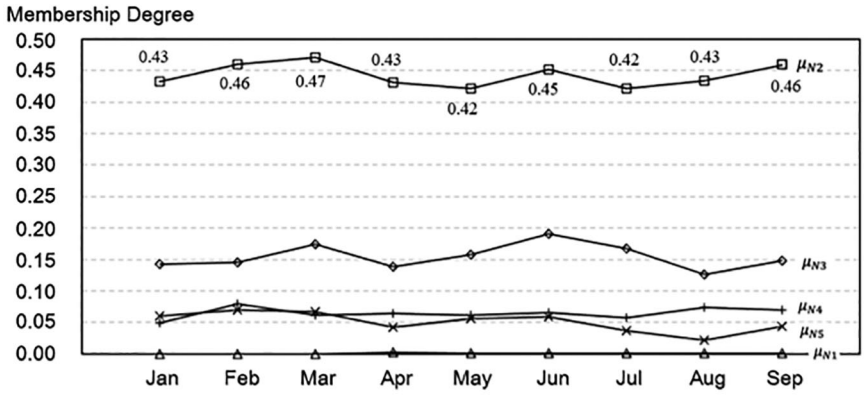


Figure 11. Evaluation results of comprehensive dam safety from January 2019 to September 2019.

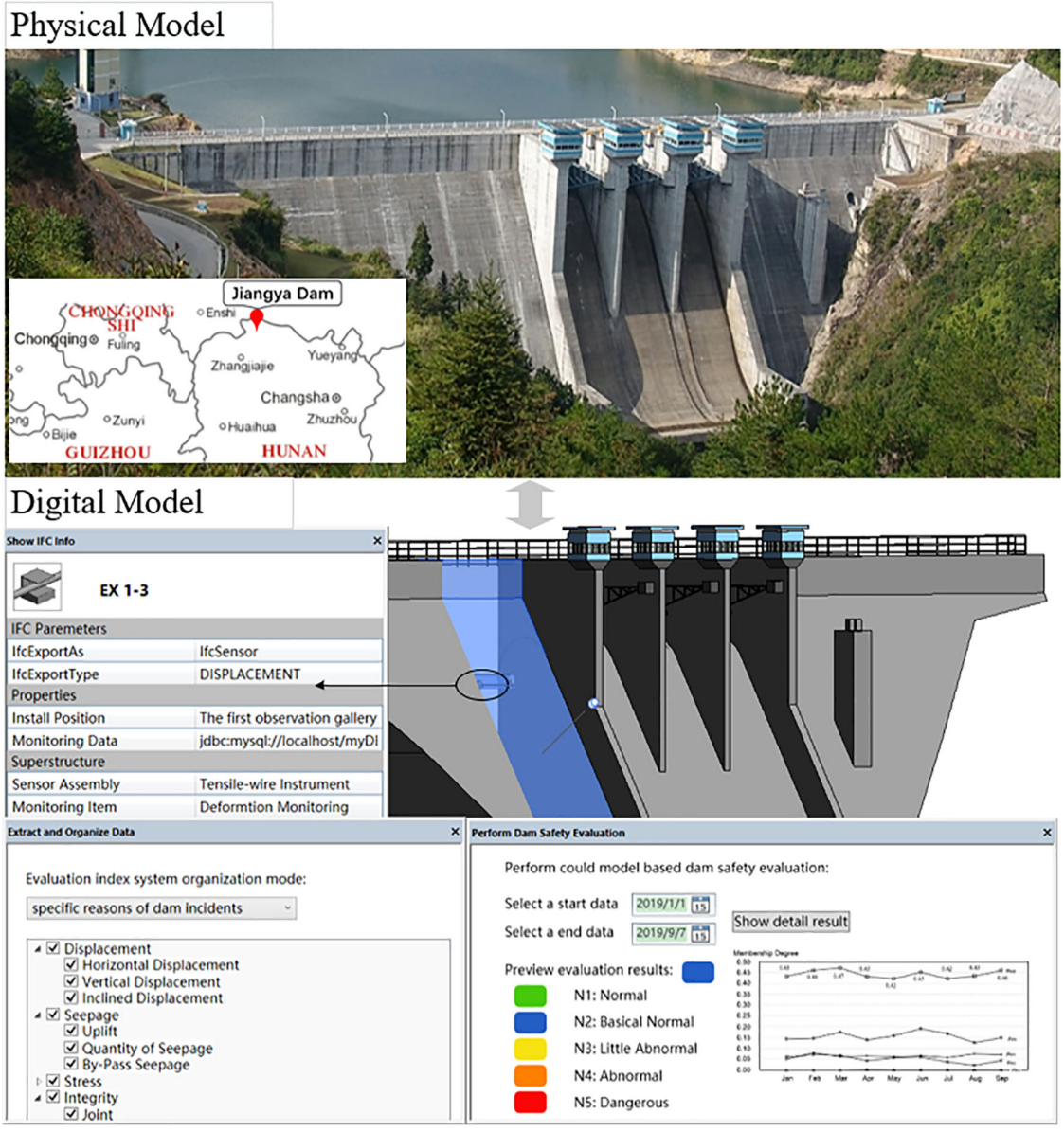


Figure 12. Dam safety evaluation powered by dam DT platform.

evaluation results of dam deformation index are listed in Table 6. In Table 6, μ_{Ni} represents the membership degree of the safety status to *i*th grade.

Finally, after all the evaluation indices in the second level are assessed, the comprehensive dam safety is obtained, the result is shown in Table 7. In this case, the weight values of

all the second-level indices are the same, that is, the deformation, seepage, stress, and integrity are equally important to the dam safety. Then, according to the maximum membership principle, the comprehensive status of the dam from September 1st to September 7th, 2019 is determined to be basically normal.

Moreover, based on the standardised information integration supported by the dam DT, a continuous evaluation of comprehensive dam safety is achieved. For instance, the evaluation results from January 2019 to September 2019 are listed in the Table 8. As shown in Figure 11, the changing trend of these membership degrees is clearly illustrated. From January 2019 to September 2019, the safety status is always the basically normal and the maximum membership degree (μ_{N2}) is fluctuating slightly around 0.45. Therefore, the dam condition is stable and there seems no sign of premature deterioration. Otherwise, according to the evaluation result, some early mitigation measures could be adopted, such as remedial treatment, operating water control, and observation frequency increase. The physical model and digital model of Jiangya dam are shown in Figure 12, the main functions implemented in this paper can be seen in the three small panels, they are "Show IFC Info", "Extract and Organise Data", and "Perform Dam Safety Evaluation", respectively. In the user interface, the results of continuous dam safety evaluation are intuitively displayed.

6. Conclusions

To achieve the dynamic and proactive mode of dam safety evaluation, a DT-enabled solution is proposed in this paper. The results of the case study verify the ability of the proposed approach to provide dynamic information and perform continuous dam safety evaluation by taking full advantage of the standardised monitoring-related information in a dam DT. It is beneficial for indicating the overall safety status, supporting decision-making, and performing preventive maintenance before severe and even catastrophic consequences happen during the O&M stage of dam projects. The main innovations of this paper are the following:

1. With the help of the standardised description of monitoring-related information in dam DT environment, the availability and interoperability of required information for complex evaluation tasks are enhanced. Compared with traditional methods of dam management, this approach can provide reliable and dynamic data to support continuous safety evaluation.
2. Based on available information and the cloud model concept, an automated evaluation approach that can consider uncertainties in the evaluation process, is proposed. This approach can present the general trend of dam safety status in a certain period. Hence, compared with the traditional static evaluation mode, it is of vital significance for diagnosing potential problems and taking reinforcement measures timely.

Although the proposed comprehensive dam evaluation approach can realise continuous monitoring of dam behaviour, as verified in the case study, there are still some shortcomings existing. For example, IFC entities related to dam structures like *IfcDam*, *IfcDamGallery*, and *IfcDamFoundation* are not included in the current IFC schema, which still need further investigation. Meanwhile, the measurement of evaluation indices is only performed based on the historical monitoring data, but the FEM-based simulation results are also useful especially when some monitoring data is missing or contaminated (Salazar et al., 2019). However, to realise the complementation of these two sources of information, further study such as the integration of simulation results into dam DTs and the ingenious combination of these two completely different kinds of information, is still required.

Disclosure statement

No potential conflict of interest was reported by the authors.

Funding

This work was supported by the National Key R&D Program of China [Grant Nos. 2018YFC1508603], the National Natural Science Foundation of China [Grant Nos. 51579086, 51739003], and the financial support from the program of China Scholarships Council [Grant Nos. CSC201906710167].

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